

HATTIESBURG-LAUREL, MS, USA

WMO: 722348

Lat: 31.467N

Lon: 89.333W

Elev: 89

StdP: 100.26

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53808

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.9	-2.4	-12.4	1.3	0.1	-9.1	1.7	2.0	8.3	12.9	7.5	11.1	1.7	0	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	35.8	24.4	34.0	24.1	32.9	24.0	26.3	32.1	25.7	31.5	25.2	31.0	2.3	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	20.0	28.8	24.0	19.1	28.2	23.6	18.6	27.8	81.8	32.0	79.2	31.7	77.3	31.0	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.3	5.4	DB	-8.0	37.3	2.5	1.5	-9.9	38.4	-11.4	39.2	-12.8	40.1	-14.7	41.2	
			WB	-9.1	27.0	2.2	0.9	-10.7	27.6	-11.9	28.2	-13.2	28.7	-14.8	29.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.8	9.0	11.3	14.8	18.5	22.8	26.1	27.3	27.2	24.9	19.3	13.5	10.3
	DBStd	7.69	5.68	5.46	4.84	3.79	2.99	1.91	1.68	1.79	2.57	4.22	4.63	5.40
	HDD10.0	229	88	44	14	1	0	0	0	0	1	17	64	
	HDD18.3	1121	291	203	128	43	6	0	0	0	1	41	154	254
	CDD10.0	3440	58	82	164	257	396	483	536	533	446	289	124	74
	CDD18.3	1291	3	8	19	49	143	233	277	274	196	71	11	6
	CDH23.3	11617	5	33	135	417	1301	2173	2627	2601	1648	598	67	12
(14) Wind	WSAvg	2.1	2.4	2.6	2.6	2.6	2.0	1.6	1.5	1.4	1.8	2.0	2.1	2.3
	PrecAvg	1506	156	137	155	127	112	127	142	119	113	81	101	135
	PrecMax	1816	310	272	338	275	280	296	250	371	283	192	234	430
	PrecMin	1075	26	29	25	21	17	15	43	35	5	2	38	38
	PrecStd	206	71	66	75	64	73	71	50	69	75	63	49	73
	DB	23.9	26.8	27.9	29.9	33.5	36.3	37.4	37.4	35.2	32.4	27.5	25.0	
	MCWB	18.0	19.3	19.1	21.1	22.8	23.7	24.4	24.1	23.8	22.1	19.6	19.4	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	22.3	24.0	27.0	28.7	32.3	34.0	35.1	35.7	33.6	30.8	25.2	22.7	
	2% MCWB	17.5	17.7	18.4	20.1	22.4	23.7	24.5	24.6	23.4	21.9	18.5	18.5	
	5% DB	20.3	22.3	24.8	27.5	31.1	32.8	33.8	33.8	32.4	28.7	23.1	21.2	
	5% MCWB	16.3	17.0	17.5	19.4	22.1	23.7	24.4	24.5	23.2	20.8	17.7	17.8	
	10% DB	18.0	20.4	22.7	26.2	29.2	32.1	32.6	32.7	31.2	27.3	22.1	19.0	
	10% MCWB	15.0	16.2	16.5	18.3	21.3	23.6	24.4	24.5	23.0	19.7	17.3	16.4	
	WB	19.9	21.2	21.2	22.8	24.7	26.4	26.9	27.3	26.1	24.9	21.9	21.2	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	22.6	23.9	25.3	27.2	30.3	31.5	32.4	32.5	32.1	30.0	24.4	23.4	
	2% WB	18.6	19.6	20.1	21.7	23.9	25.5	26.1	26.3	25.3	23.8	20.7	19.6	
	2% MCDB	21.1	22.7	23.9	26.3	29.3	31.0	32.2	32.0	30.5	28.0	23.4	22.1	
	5% WB	17.3	18.3	19.1	20.9	23.2	24.9	25.6	25.7	24.6	22.7	19.4	18.3	
	5% MCDB	19.2	21.3	23.0	25.0	28.4	30.6	31.7	31.5	29.7	26.3	22.4	20.5	
	10% WB	15.6	16.8	18.0	20.0	22.4	24.4	25.1	25.2	23.9	21.6	18.0	16.8	
	10% MCDB	18.0	19.5	21.8	24.3	27.7	29.9	31.0	31.0	28.7	24.8	20.8	18.7	
(35) Mean Daily Temperature Range	MDBR	12.0	12.1	12.7	12.9	12.3	11.2	10.9	11.1	11.6	12.9	13.0	11.3	
	5% DB	12.2	12.6	13.5	13.0	13.5	12.5	12.5	12.8	13.1	13.7	13.9	11.7	
	5% WB	8.6	7.8	7.0	6.5	5.6	4.9	4.7	4.8	5.2	6.4	8.6	8.4	
	MCDBR	10.0	10.7	11.0	10.5	11.2	10.8	11.0	11.2	10.7	10.3	11.1	10.2	
	MCWBR	8.0	7.4	6.7	6.1	5.2	4.9	4.8	4.8	4.6	5.7	7.9	8.1	
(40) Clear-Sky Solar Irradiance	taub	0.326	0.339	0.365	0.398	0.431	0.454	0.514	0.502	0.445	0.375	0.346	0.329	
	taud	2.526	2.486	2.422	2.336	2.277	2.258	2.111	2.143	2.275	2.444	2.472	2.528	
	Ebn at Noon	902	921	917	894	863	838	787	791	826	869	868	878	
	Edh at Noon	86	98	112	127	136	138	159	152	128	101	90	82	
(44) All-Sky Solar Radiation	RadAvg	2.81	3.37	4.47	5.59	6.11	6.14	5.88	5.60	5.06	4.31	3.24	2.48	
	RadStd	0.25	0.48	0.36	0.45	0.37	0.46	0.34	0.38	0.36	0.53	0.30	0.25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	+85

Nomenclature: See separate page